

QT 406: Introduction to Quantum Communication

Course Code	QT 406
Course Name	Introduction to Quantum Communication
L-T-P-C	3-0-0-3
Prerequisites	Engineering Mathematics (Linear Algebra, Complex algebra, basics of 2nd of ODEs and initial value problems, 2nd order PDEs and boundary value problems, Probability and Statistics, Random variables). Maxwell's equations and EM theory at the level of the core physics syllabus from AICTE model
Intended for	UG/PG/PhD
Distribution	Elective PG/Elective UG
Semester	Odd/Even
Approval	47th Senate (Sep 2025)

Preamble

Students of this course learn (i) The basics of EM theory, (ii) The basics of photodetection, (iii) The basics of information theory, (iv) The central ideas in quantum communications

Course Contents

- **Basics of Polarization optics**
 - Quarter and half-wave plates
 - Polarizing beam splitters
- **Basics of linear and square-law detectors**
 - Quadrature amplitude modulation
 - Heterodyne and Homodyne demodulation and linear detectors
 - Intensity measurements and square law detectors
 - Photomultipliers, Avalanche Photo diodes
- **Digital communication – information theory (basics)**
 - Information entropy
 - Noiseless channel encoding
 - Noisy channel encoding
- **No cloning theorem**
- **Quantum Memories**
- **Quantum repeaters**
- **Entanglement and Bell Theorems**
- **Bell Measurements and Tests**
- **Quantum Teleportation protocol**
- **Quantum Dense coding**
- **Quantum Key Distribution protocols**
 - BB84
 - E91
 - BBM92
 - B92
 - COW
 - DPS
- **Quantum Networks and Quantum Internet**
- **Survey of Hardware implementations**
 - Free space communications
 - Satellite based communications
 - Fibre optics-based communications

References

1. *Quantum computation and quantum information* – Nielsen and Chuang Cambridge University Press, Cambridge (2010)
2. A Pathak, *Elements of Quantum Computation and Quantum Communication*, Boca Raton, CRC Press (2015)